

When Could Relativity Break a Raft Lease?

Orders of Magnitude

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Independent study · technical note to the survey “Raft under Physical Communication Models”

Abstract. The survey found one Raft mechanism whose safety rests on clocks, the leader lease, and no frame-invariant statement of its safety condition. Before proving such a statement, this note asks which physical effects could matter and how large they are. Which condition counts as “safe” depends on the correctness notion. If a stale read only has to be causally undetectable, the lease gains a full round trip of slack. Gravitational and motional rate offsets of order 10^{-10} could then exhaust that slack only after election timeouts of months to millennia; every realistic configuration is safe by several orders of magnitude. If the read must be correct in every reference frame, the notion Raft itself satisfies, the slack disappears and the lease must shrink by the Doppler factor between leader and follower. That effect is first order in $\beta = v/c$, between 10^{-6} and 10^{-4} for satellite and planetary links. On those links it exceeds the drift budget of every oscillator from an oven-controlled crystal upward, and on low-orbit and Mars links even that of a temperature-compensated crystal. A numerical model in flat 1+1D spacetime supports both statements; proofs are left to the next step.

1 Question and setup

Ongaro’s lease rule reads: once a majority acknowledges a round of heartbeats, the leader extends its lease to “start + election timeout / clock drift bound”, where *start* is when it sent the round, because followers ignore vote requests for an election timeout after hearing from the leader [1, p. 92]. The rule assumes that no server’s clock advances more than the drift bound times any other’s over a given period. Comparing distant clocks over “the same” period presupposes a simultaneity convention, which is the gap the survey identified.

The lease round can be stated with events instead (Figure 1). The leader sends the heartbeat round at S and starts its lease there. A follower receives it at R and refuses to vote for T_e of its own clock, until V . Light from V reaches the leader at A . The lease expires at E , T_L of leader time after S . Write D for the last leader event whose light reaches V .

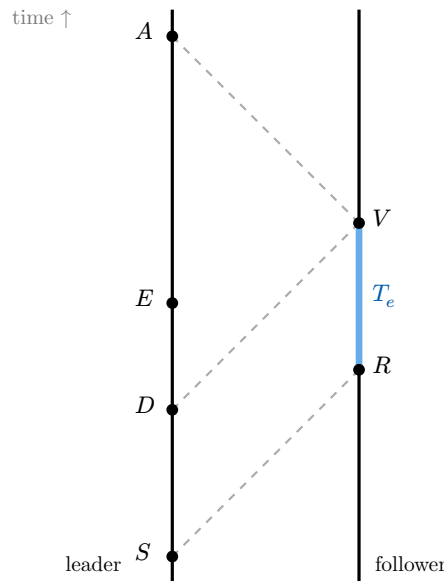


Figure 1: One lease round for two nodes at rest, light signals dashed. The follower’s no-vote window (blue) runs from R to V . Under the causal reading the lease may last until just before A ; under the every-frame reading only until D .

The expiry E as drawn is safe under the first reading and not under the second.

A new leader needs a vote from some follower that acknowledged the round, so its first write completes at an event in the causal future of that follower’s V . Assume the worst case, where a new majority and its client sit next to the follower, so the write can complete arbitrarily soon after V . A read served by the old leader at E is stale with respect to that write. Whether this violates correctness depends on which relativistic linearizability is required [2]:

- **Causal reading.** A stale read violates correctness only if the write’s completion is in the causal past of the read. The pairwise condition is that E is not in the causal future of V , so E must come before A . This is the condition behind R1-linearizability (linearizable in some frame).
- **Every-frame reading.** R2- and R3-linearizability require linearizability in every total order that extends causality. For two spacelike-separated events, some such order puts either one first, so the read must causally precede the write. The pairwise condition is that E is in the causal past of V : E must come no later than D . Raft without leases is R3-linearizable [3], so this is the reading under which leases keep Raft’s guarantee.

Both are necessary pairwise conditions, not proofs of R1 or R3 for whole histories.

2 Clock rates

To first order in the weak field, a clock moving at speed v at Newtonian potential Φ ticks at $d\tau/dt = 1 + \Phi/c^2 - v^2/(2c^2)$ relative to the coordinate time of the chosen frame [4]. In a circular orbit of radius r this gives $-3GM/(2rc^2)$; a clock on the rotating geoid runs at $-L_G$ relative to geocentric coordinate time, with the defining constant L_G and all mass parameters from the IERS Conventions [5]. Planetary dimensions come from NASA’s fact sheets [6], [7]. The model neglects higher multipoles, orbital eccentricity, Shapiro delay and tidal terms. Table 1 checks it against published values.

Table 1: Validation of the rate model. The Mars difference comes from the circular-orbit approximation; the published value varies by ± 226 $\mu\text{s/day}$ over a Mars year.

Clock pair	This model	Published
GPS orbit vs. geoid, fractional	4.4647×10^{-10}	4.4647×10^{-10} [4]
Moon surface vs. Earth, $\mu\text{s/day}$	56.00	56.02 [8]
Mars surface vs. Earth, $\mu\text{s/day}$, mean	488	477 ± 226 [9]

3 Lease terms in realistic configurations

Table 2 lists seven configurations. Distances and the largest range rate $\beta_{\max}c$ come from circular coplanar orbits sampled over one synodic period, counting only times when the two nodes can see each other. The crossing LEO pair is modelled as two counter-rotating orbits; ground–GPS geometry is equatorial and understates the range rate of the real inclined orbits. The election timeout T_e is the smallest that satisfies Raft’s timing requirement: about ten times the worst one-way latency and at least 150 ms [1, pp. 150, 152].

Table 2: Configurations: two servers on floors 10 m apart; two ground stations 1000 km apart at heights differing by 1 km; a ground station and a satellite at 400 km; satellites at 400 km and 550 km on crossing orbits; a ground station and a GPS satellite; Earth and Moon surfaces; Earth and Mars surfaces. δ is the fractional rate offset between the two clocks; T_e^* is the election timeout at which the rate term would use up the round-trip slack of the causal reading.

Configuration	$d_{\min}$ (m)	$d_{\max}$ (m)	$\beta_{\max}$	δ	T_e (s)	T_e^* (years)
Floors	10.0	10.0	0	1.09×10^{-15}	0.150	1.94
Ground-ground	1.00×10^6	1.00×10^6	0	1.09×10^{-13}	0.150	1943
Ground-LEO	4.00×10^5	2.29×10^6	2.25×10^{-5}	-2.85×10^{-10}	0.150	0.297
LEO-LEO	1.50×10^5	5.00×10^6	5.03×10^{-5}	2.12×10^{-11}	0.167	1.49
Ground-GPS	2.02×10^7	2.58×10^7	1.55×10^{-6}	4.46×10^{-10}	0.860	9.56
Earth-Moon	3.78×10^8	3.84×10^8	1.49×10^{-6}	6.48×10^{-10}	12.8	123
Earth-Mars	7.84×10^{10}	3.78×10^{11}	4.65×10^{-5}	5.65×10^{-9}	1.26×10^4	2934

Causal reading. To first order the leader’s proper time from S to A is $T_e(1 - \delta) + \text{RTT}$, where δ is the follower’s rate relative to the leader’s. The lease can be unsafe only if $|\delta| T_e > \text{RTT}$, that is for

$T_e > T_e^* = \text{RTT}/|\delta|$. Across the table T_e^* is at least 0.297 years, against election timeouts of at most 1.26×10^4 s. Under this reading relativity never threatens a lease that respects Raft’s own timing requirement, and Ongaro’s rule needs no change.

Every-frame reading. Here the slack is gone: the follower receives the leader’s signals stretched by the Doppler factor k , so the latest safe expiry D comes T_e/k of leader time after S . For an inertial pair $k = \sqrt{(1+\beta)/(1-\beta)}$ [10], and in general $k_{\max} - 1 \approx \beta_{\max} + |\delta|$ to first order. Ongaro’s rule spends a drift budget of about $2\rho T_e$, where ρ is the clock’s rate tolerance. Table 3 compares the two using oscillator accuracies from Vig’s tutorial [11, slide 2-8], including environment and one year of aging. A share above 1 means the lease is unsafe under this reading even with perfect clocks.

Table 3: Share of the classical drift budget consumed by the Doppler factor, $(k_{\max} - 1)/(2\rho)$. The last column is how long a TCXO-timed lease outlives the every-frame bound with perfect clocks, $(k_{\max} - 1 - 2\rho)T_e$.

Configuration	XO	TCXO	OCXO	Rb	Cs	TCXO window (s)
ρ	1×10^{-4}	1×10^{-6}	1×10^{-8}	1×10^{-9}	1×10^{-11}	
Floors	5.44×10^{-12}	5.44×10^{-10}	5.44×10^{-8}	5.44×10^{-7}	5.44×10^{-5}	0
Ground-ground	5.44×10^{-10}	5.44×10^{-8}	5.44×10^{-6}	5.44×10^{-5}	5.44×10^{-3}	0
Ground-LEO	0.113	11.3	1126	1.13×10^4	1.13×10^6	3.08×10^{-6}
LEO-LEO	0.252	25.2	2517	2.52×10^4	2.52×10^6	8.06×10^{-6}
Ground-GPS	7.76×10^{-3}	0.776	77.6	776	7.76×10^4	0
Earth-Moon	7.48×10^{-3}	0.748	74.8	748	7.48×10^4	0
Earth-Mars	0.233	23.3	2327	2.33×10^4	2.33×10^6	0.561

Doppler consumes up to a quarter of a commodity quartz budget on the fastest links, exceeds the whole budget of a temperature-compensated oscillator for LEO and Mars links, and dominates every better clock by orders of magnitude. The exposure windows are short, microseconds for Earth-orbit links, but reach 0.561 s for an Earth–Mars lease. A cluster with its leader on Earth and a majority on Mars could elect a new leader within such a window.

4 Numerical test of the survey’s conjecture

The survey conjectured that an inertial leader cannot lose lease time, because by the reverse triangle inequality its proper time from S to A is at least T_e , so that first-order Doppler effects cancel. A small model of flat 1+1D spacetime with $c = 1$ (`analysis/spacetime.py`) tests this. It represents worldlines as inertial, piecewise inertial or uniformly accelerated, and finds light-signal events by root finding.

- In 2000 random lease rounds with an inertial leader and a piecewise-inertial follower at speeds up to $0.9c$, the smallest ratio of the causal-reading lease bound to T_e was 1.078. It was never below 1.
- A leader that flies out at $0.8c$ and returns to a resting follower accumulates 6.0 units of proper time while the follower’s $T_e = 10.0$ elapses: the twin paradox, and a counterexample for non-inertial leaders.
- For a receding inertial pair the every-frame bound equals T_e/k : Ground-LEO, $\beta = 2.25 \times 10^{-5}$: $0.9999775 T_e$; Earth-Mars, $\beta = 4.65 \times 10^{-5}$: $0.9999535 T_e$; Fast pair, $\beta = 0.600$: $0.5000000 T_e$.

The conjecture therefore holds, so far numerically, for the causal reading only. Under the every-frame reading first-order Doppler does not cancel. That corrects the survey’s remark that “first-order Doppler effects cancel”: they cancel only when correctness is required in some frame, not in every frame.

5 What Step 2 must prove

Measuring each clock’s drift against its own proper time makes the drift bound frame-invariant. With that, two statements survive the tests above:

1. **Causal reading.** In 1+1D Minkowski spacetime, with an inertial leader and a piecewise-inertial follower whose clocks stay within ρ of proper time, the lease $T_L \leq T_e(1 - \rho)/(1 + \rho)$ ends before A .
2. **Every-frame reading.** With a Doppler bound $k_{\max}$ between leader and follower, the lease $T_L \leq T_e(1 - \rho)/((1 + \rho)k_{\max})$ ends no later than D .

The first is the reverse triangle inequality applied to the chain $S \rightarrow R \rightarrow V \rightarrow A$. The second needs only the Doppler factor, which the follower could measure by comparing heartbeat spacing with its own clock.

6 Limitations

The rates are weak-field and first order, and the orbits are circular and coplanar. The lease conditions are pairwise and assume the fastest possible new leader. The spacetime model is flat and one-dimensional, so gravity enters only through the rate offsets of Table 2. The numerical tests are evidence, not proofs.

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